MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, HEX BUS DRIVERS WITH 3-STATE OUTPUTS, MONOLITHIC SILICON

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, hex bus driver (3-state) microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Hex bus driver, gated enable inputs for X-Y coincident bus control
02	Hex inverter bus driver, gated enable inputs for X-Y coincident bus control
03	Hex bus driver, 4-line and 2-line enable inputs
04	Hex inverter bus driver, 4-line and 2-line enable inputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -12 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) per device 1/	
Device types 01 and 03	275 mW dc
Device type 02	550 mW dc
Device type 04	165 mW dc
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases E and F	(See appendix C of MIL-M-38510)
Junction temperature (T_J) 2/	+175°C

1/ Must withstand the added P_D due to short circuit condition (e.g. I_{OS}) at one output for five seconds duration.
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC}) - - - - - - - - - - - - - - - -	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH}) - - - - - -	2.0 V dc
Maximum low-level input voltage (V_{IL})- - - - - -	0.8 V dc
Normalized fanout (each input) 3/- - - - - - - -	20 maximum
Case operating temperature range (T_C)- - - - - -	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Case outlines. Case outlines shall be as specified in 1.2.3.

3.2.2 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figure 1.

3.2.3 Truth tables. The truth tables shall be as specified on figure 2.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I and apply over the full recommended case operating temperature range.

3/ The device shall fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IN} = 0.8 V or 2.0 V, I_{OH} = -2 mA	All	2.4		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IN} = 2.0 V or 0.8 V, I_{OL} = 32 mA	All		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -12 mA, T_C = 25°C	All		-1.5	V
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	All		40	µA
High-level input current	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	All		1	mA
Output clamp voltage	V_{OC}	V_{CC} = 5.5 V, I_{OC} = -12 mA, T_C = 25°C	All		-1.5	V
Inhibited-state output leakage current	I_{O1}	V_{CC} = 5.5 V, V_{IN} = 2.0 V or 0.8 V, V_{OUT} = 2.4 V	All		40	µA
	I_{O2}	V_{CC} = 5.5 V, V_{IN} = 2.0 V or 0.8 V, V_{OUT} = 0.4 V	All		-40	µA
Low-level input current at A input	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.5 V, Both $\overline{G}$ inputs at 2.0 V	All		-40	µA
Low-level input current at A input	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V, Both $\overline{G}$ inputs at 0.4 V	All		-1.6	mA
Low-level input current at $\overline{G}$ input	I_{IL3}	V_{CC} = 5.5 V, V_{IN} = 0.4 V,	All		-1.6	mA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V 1/	All	-40	-130	mA
Supply current	I_{CC}	V_{CC} = 5.5 V	01 02 03 04		85 77 85 77	mA
Propagation delay time (low-to-high level output)	t_{PLH}	V_{CC} = 5.0 V, R_L = 400Ω, C_L = 50 pF	01 02 03 04	2 2 2 2	21 22 21 22	ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
Propagation delay time (high-to-low level output)	t_{PHL}	V_{CC} = 5.0 V, R_L = 400Ω, C_L = 50 pF	01 02 03 04	2 2 2 2	27 21 27 21	ns
Propagation delay time (disabled to high-level output)	t_{PZH}	V_{CC} = 5.0 V, R_L = 400Ω, C_L = 50 pF	01 02 03 04	2 2 2 2	40 40 40 40	ns
Propagation delay time (high-level to disabled output)	t_{PHZ}	V_{CC} = 5.0 V, R_L = (see figure 4), C_L = 50 pF	01 02 03 04	2 2 2 2	20 20 20 20	ns
Propagation delay time (disabled to low-level output)	t_{PZL}	V_{CC} = 5.0 V, R_L = 400Ω, C_L = 50 pF	01 02 03 04	2 2 2 2	42 42 42 42	ns
Propagation delay time (low-level to disabled output)	t_{PLZ}	V_{CC} = 5.0 V, R_L = (see figure 4), C_L = 50 pF	01 02 03 04	2 2 2 2	32 32 32 32	ns

1/ Not more than one output should be shorted at a time.

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3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,9, 10,11	1*,2,3,9
Group A test requirements (method 5005)	1,2,3,9, 10,11	1,2,3,9, 10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,9, 10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	None
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 2 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions, or equivalent:

(1) Test condition D, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

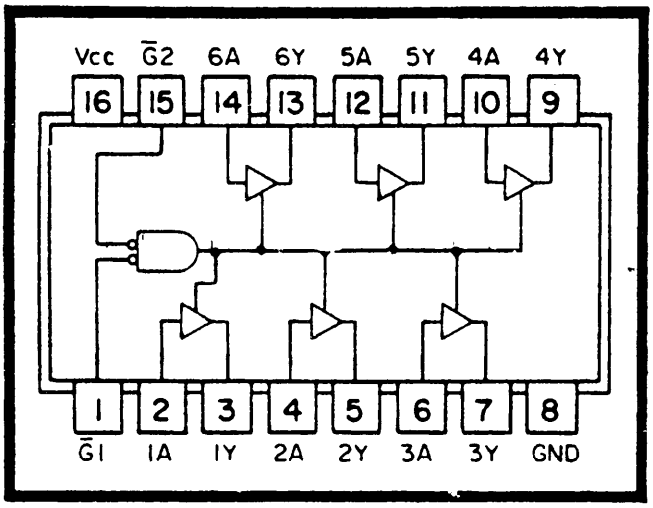
a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

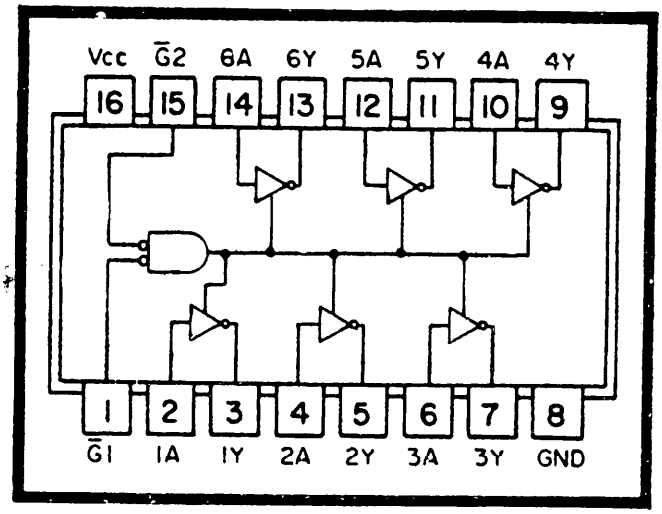
c. Requirement for certificate of compliance, if applicable.

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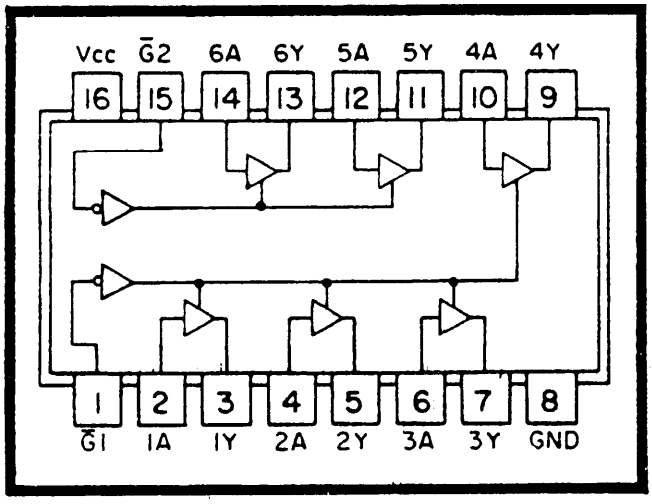
Device type 01



Device type 02



Device type 03



Device type 04

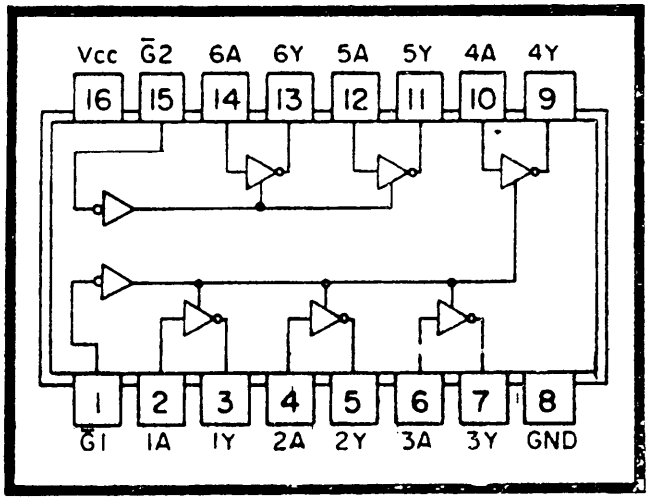


FIGURE 1. Terminal connections and logic diagrams.

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Device type 01

INPUTS			OUTPUT
$\overline{G1}$	$\overline{G2}$	A	Y
H	X	X	Z
X	H	X	Z
L	L	H	H
L	L	L	L

Device type 02

INPUTS			OUTPUT
$\overline{G1}$	$\overline{G2}$	A	Y
H	X	X	Z
X	H	X	Z
L	L	H	L
L	L	L	H

Device type 03

INPUTS		OUTPUT
$\overline{G}$	A	Y
H	X	Z
L	H	H
L	L	L

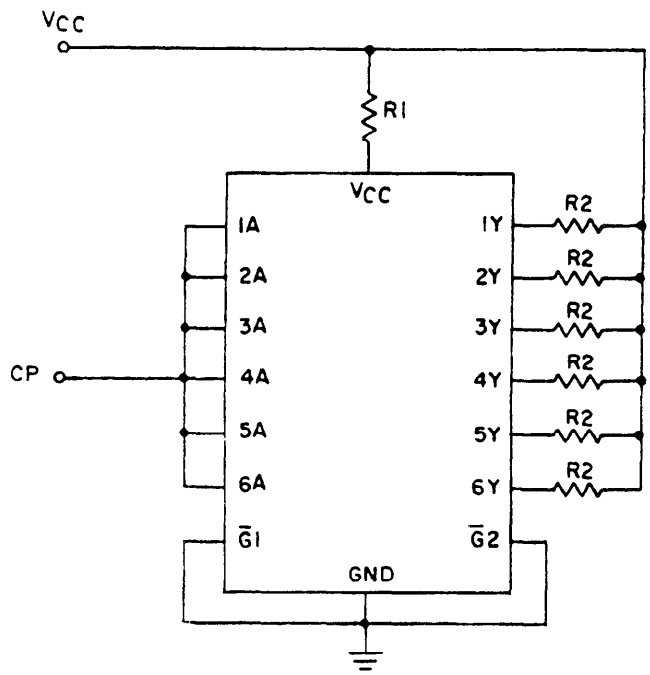
Device type 04

INPUTS		OUTPUT
$\overline{G}$	A	Y
H	X	Z
L	H	L
L	L	H

H = high level, L = low level,
X = irrelevant, Z = high-impedance

FIGURE 2. Truth tables (each driver).

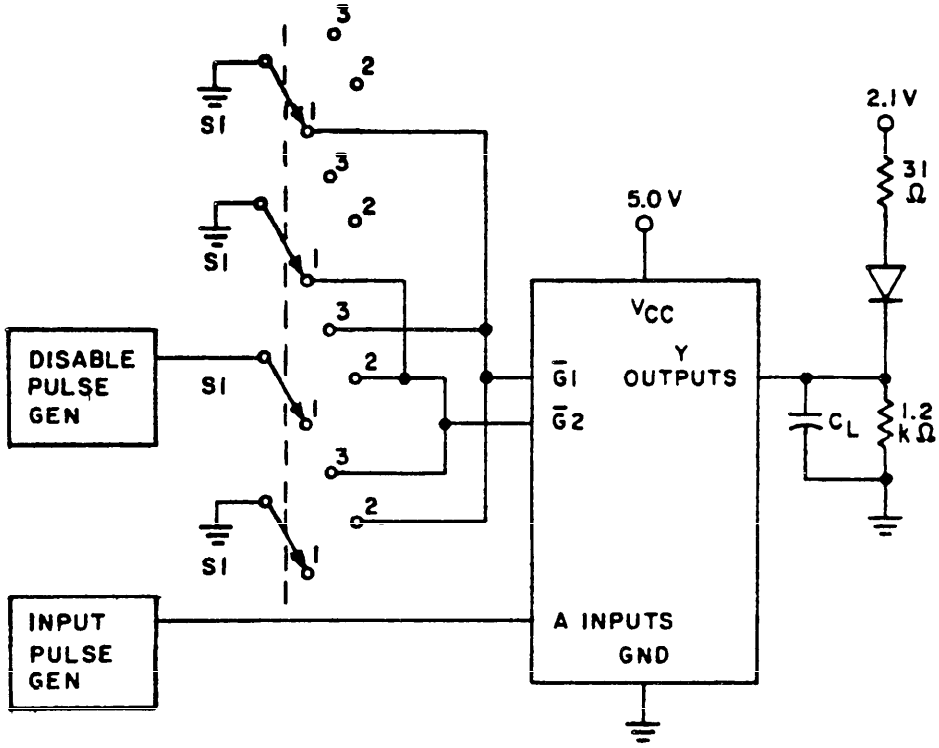
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NOTES:
1. V_{CC} and R1 are such that the minimum voltage at the device terminal is 5.0 volts.
2. R2 = 140Ω ±5%.
3. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V to 0.7 V; V_{IH} = 2.0 V to 5.5 V.

FIGURE 3. Burn-in and life test circuits.

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	S1	C_L
t_{PLH}	Pos. 1	50 pF
t_{PHL}	Pos. 1	50 pF
t_{PHZ}	Pos. 2 or 3	50 pF
t_{PLZ}	Pos. 2 or 3	50 pF
t_{PZL}	Pos. 2 or 3	50 pF
t_{PZH}	Pos. 2 or 3	50 pF

FIGURE 4. Switching time test circuit.

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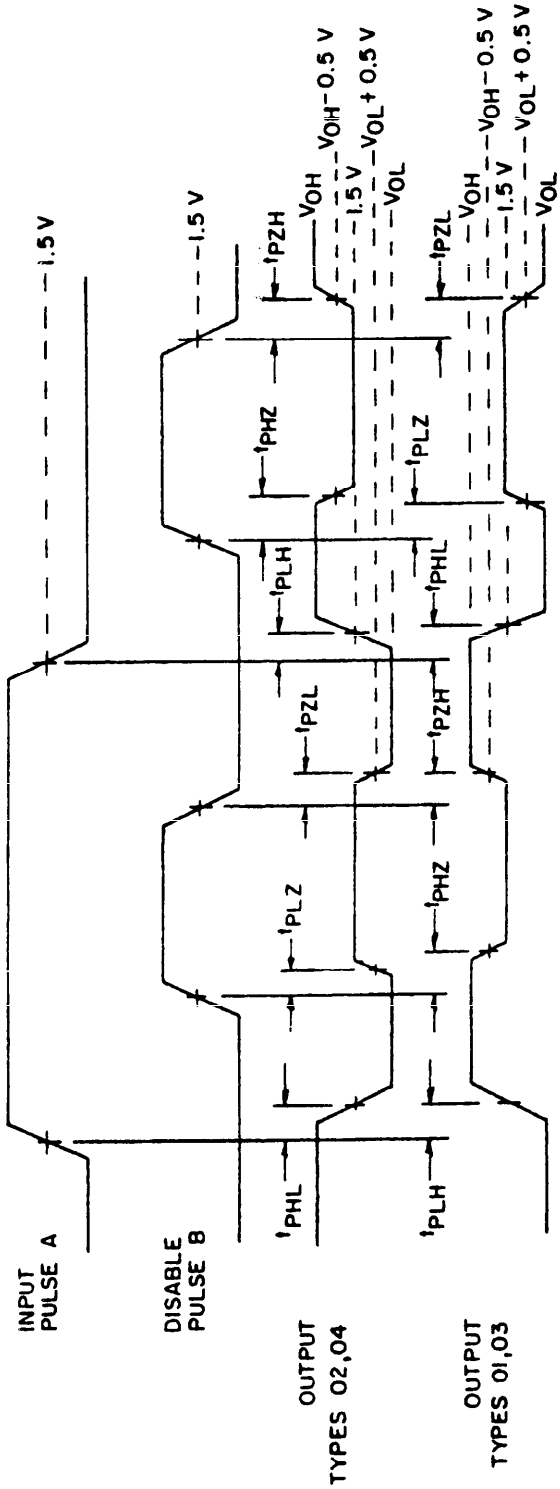


FIGURE 4. Switching time test circuit - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{G}_1$	1_A	1_Y	2_A	2_Y	3_A	3_Y	GND	4_Y	4_A	5_Y	5_A	6_Y	6_A	$\overline{G}_2$	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	0.8 V	2.0 V	-2 mA					GND							0.8 V	4.5 V	1_Y	2.4		V
		"	2	"			2.0 V	-2 mA	2.0 V	-2 mA	"	-2 mA	2.0 V	-2 mA	2.0 V	-2 mA	2.0 V	"	"	2_Y	"		"
		"	3	"							"							"	"	3_Y	"		"
		"	4	"							"							"	"	4_Y	"		"
		"	5	"							"							"	"	5_Y	"		"
		"	6	"							"							"	"	6_Y	"		"
	V_{OL}	3007	7	"	0.8 V	32 mA					"							"	"	1_Y		0.4	"
		"	8	"			0.8 V	32 mA	0.8 V	32 mA	"	32 mA	0.8 V	32 mA	0.8 V	32 mA	0.8 V	"	"	2_Y	"	"	"
		"	9	"							"							"	"	3_Y	"	"	"
		"	10	"							"							"	"	4_Y	"	"	"
		"	11	"							"							"	"	5_Y	"	"	"
		"	12	"							"							"	"	6_Y	"	"	"
	I_{O1}		13	2.0 V	2.0 V	2.4 V					"							"	5.5 V	1_Y		40	µA
			14	0.8 V	0.8 V	2.4 V					"							2.0 V	"	1_Y		"	"
			15	2.0 V			2.0 V	2.4 V			"							0.8 V	"	2_Y		"	"
			16	0.8 V			0.8 V	2.4 V			"							2.0 V	"	2_Y		"	"
			17	2.0 V					2.0 V	2.4 V	"							0.8 V	"	3_Y		"	"
			18	0.8 V					0.8 V	2.4 V	"							2.0 V	"	3_Y		"	"
			19	2.0 V							"	2.4 V	2.0 V					0.8 V	"	4_Y		"	"
			20	0.8 V							"	2.4 V	0.8 V					2.0 V	"	4_Y		"	"
			21	2.0 V							"			2.4 V	2.0 V			0.8 V	"	5_Y		"	"
			22	0.8 V							"			2.4 V	0.8 V			2.0 V	"	5_Y		"	"
			23	2.0 V							"					2.4 V	2.0 V	0.8 V	"	6_Y		"	"
			24	0.8 V							"					2.4 V	0.8 V	2.0 V	"	6_Y		"	"
	I_{O2}		25	2.0 V	0.8 V	0.4 V					"							0.8 V	"	1_Y		-40	"
			26	0.8 V	2.0 V	0.4 V					"							2.0 V	"	1_Y		"	"
			27	2.0 V			0.8 V	0.4 V			"							0.8 V	"	2_Y		"	"
			28	0.8 V			2.0 V	0.4 V			"							2.0 V	"	2_Y		"	"
			29	2.0 V					0.8 V	0.4 V	"							0.8 V	"	3_Y		"	"
			30	0.8 V					2.0 V	0.4 V	"							2.0 V	"	3_Y		"	"
			31	2.0 V							"	0.4 V	0.8 V					0.8 V	"	4_Y		"	"
			32	0.8 V							"	0.4 V	2.0 V					2.0 V	"	4_Y		"	"
			33	2.0 V							"			0.4 V	0.8 V			0.8 V	"	5_Y		"	"
			34	0.8 V							"			0.4 V	2.0 V			2.0 V	"	5_Y		"	"
			35	2.0 V							"					0.4 V	0.8 V	0.8 V	"	6_Y		"	"
			36	0.8 V							"					0.4 V	2.0 V	2.0 V	"	6_Y		"	"
	V_{IC}		37		-12 mA						"								4.5 V	1_A		-1.5	V
			38				-12 mA				"								"	2_A		"	"
			39						-12 mA		"								"	3_A		"	"
			40								"		-12 mA						"	4_A		"	"
			41								"				-12 mA				"	5_A		"	"
			42								"						-12 mA		"	6_A		"	"
			43	-12 mA							"								"	$\overline{G}_1$		"	"
			44								"							-12 mA	"	$\overline{G}_2$		"	"
	I_{IH1}	3010	45	2.4 V							"								5.5 V	$\overline{G}_1$		40	µA
		"	46		2.4 V						"								"	1_A		"	"
		"	47				2.4 V				"								"	2_A		"	"
		"	48						2.4 V		"								"	3_A		"	"
		"	49								"		2.4 V						"	4_A		"	"
		"	50								"				2.4 V				"	5_A		"	"
		"	51								"						2.4 V		"	6_A		"	"
		"	52								"							2.4 V	"	$\overline{G}_2$		"	"

See foot[illegible] at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{G}_1$	1_A	1_Y	2_A	2_Y	3_A	3_Y	GND	4_Y	4_A	5_Y	5_A	6_Y	6_A	$\overline{G}_2$	V_{CC}				
1 T_C = +25°C	I_{IH2}	3010	53	5.5 V							GND								5.5 V	$\overline{G}_1$		1	mA
		"	54		5.5 V						"								"	1_A		"	"
		"	55				5.5 V				"								"	2_A		"	"
		"	56						5.5 V		"								"	3_A		"	"
		"	57								"		5.5 V						"	4_A		"	"
		"	58								"				5.5 V				"	5_A		"	"
		"	59								"						5.5 V		"	6_A		"	"
		"	60								"							5.5 V	"	$\overline{G}_2$		"	"
	I_{IL1}	3009	61	2.0 V	0.5 V						"							2.0 V	"	1_A		-40	μA
		"	62	"			0.5 V				"							"	"	2_A		"	"
		"	63	"					0.5 V		"							"	"	3_A		"	"
		"	64	"							"		0.5 V					"	"	4_A		"	"
		"	65	"							"				0.5 V			"	"	5_A		"	"
		"	66	"							"						0.5 V	"	"	6_A		"	"
	I_{IL2}	"	67	0.4 V	0.4 V						"							0.4 V	"	1_A		-1.6	mA
		"	68	"			0.4 V				"							"	"	2_A		"	"
		"	69	"					0.4 V		"							"	"	3_A		"	"
		"	70	"							"		0.4 V					"	"	4_A		"	"
		"	71	"							"				0.4 V			"	"	5_A		"	"
		"	72	"							"						0.4 V	"	"	6_A		"	"
	I_{IL3} I_{IL3}	"	73 74	"							" "							0.4 V	" "	$\overline{G}_1$ $\overline{G}_2$		" "	" "
	V_{OC}		75			-12 mA		-12 mA		-12 mA	"	-12 mA		-12 mA		-12 mA			"	1_Y		-1.5	V
			76								"								"	2_Y		"	"
			77								"								"	3_Y		"	"
			78								"								"	4_Y		"	"
			79								"								"	5_Y		"	"
			80								"								"	6_Y		"	"
	I_{OS}	3011	81	0.8 V	5.5 V	GND	5.5 V	GND	5.5 V	GND	"	GND	5.5 V	GND	5.5 V	GND	5.5 V	0.8 V	"	1_Y	-40	-130	mA
		"	82	"							"							"	"	2_Y	"	"	"
		"	83	"							"							"	"	3_Y	"	"	"
		"	84	"							"							"	"	4_Y	"	"	"
		"	85	"							"							"	"	5_Y	"	"	"
		"	86	"							"							"	"	6_Y	"	"	"
	I_{CC}	3005	87	GND	GND		GND		GND		"		GND		GND		GND	GND	"	V_{CC}		85	"
2	Same tests, terminal conditions and limits as for subgroup 1, except V_{IC} and V_{OC} tests are omitted and T_C = +125°C.																						
3	Same tests, terminal conditions and limits as for subgroup 2, except T_C = -55°C.																						
9 T_C = +25°C	t_{PLH}	3003 (Fig. 4)	88	GND	A	OUT	A	OUT	A	OUT	GND	OUT	A	OUT	A	OUT	A	GND	5.0 V	1_A to 1_Y	2	16	ns
		"	89	"							"							"	"	2_A to 2_Y	"	"	"
		"	90	"							"							"	"	3_A to 3_Y	"	"	"
		"	91	"							"							"	"	4_A to 4_Y	"	"	"
		"	92	"							"							"	"	5_A to 5_Y	"	"	"
		"	93	"							"							"	"	6_A to 6_Y	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1 $\overline{G}_1$	2 1_A	3 1_Y	4 2_A	5 2_Y	6 3_A	7 3_Y	8 GND	9 4_Y	10 4_A	11 5_Y	12 5_A	13 6_Y	14 6_A	15 $\overline{G}_2$	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9	t_{PHL}	3003	108	GND	A	OUT	A	OUT	A	OUT	GND	OUT	A	OUT	A	OUT	A	GND	5.0 V	1_A to 1_Y	0	25	ns
	"	"	111	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	2_A to 2_Y	"	"	"
	"	"	112	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	3_A to 3_Y	"	"	"
	"	"	113	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	4_A to 4_Y	"	"	"
	"	"	114	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	5_A to 5_Y	"	"	"
	"	"	115	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	6_A to 6_Y	"	"	"
	t_{PZH}	"	116	B	2.0 V	OUT	2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V	"	"	$\overline{G}_1$ to 1_Y	"	40	"
	t_{PZL}	"	117	GND	0.8 V	"	0.8 V		0.8 V		"		0.8 V		0.8 V		0.8 V	B	"	$\overline{G}_2$ to 1_Y	"	42	"
	t_{PHZ}	"	118	B	2.0 V	"	2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V	GND	"	$\overline{G}_1$ to 1_Y	"	20	"
	t_{PLZ}	"	119	B	0.8 V	"	0.8 V		0.8 V		"		0.8 V		0.8 V		0.8 V	GND	"	$\overline{G}_1$ to 1_Y	"	32	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

A = Pulse from input pulse gen.

B = Pulse from disable gen.

			42	-12 mA							"						-12 mA	"	$\overline{G}_2$		"	"
	I_{IH1}	3010	43	2.4 V	2.4 V		2.4 V		2.4 V		"		2.4 V		2.4 V		2.4 V	5.5 V	$\overline{G}_1$		40	µA
	"	"	44	"	"		"		"		"		"		"		"	"	1_A		"	"
	"	"	45	"	"		"		"		"		"		"		"	"	2_A		"	"
	"	"	46	"	"		"		"		"		"		"		"	"	3_A		"	"
	"	"	47	"	"		"		"		"		"		"		"	"	4_A		"	"
	"	"	48	"	"		"		"		"		"		"		"	"	5_A		"	"
	"	"	49	"	"		"		"		"		"		"		"	"	6_A		"	"
	"	"	50	"	"		"		"		"		"		"		"	"	$\overline{G}_2$		"	"
	"	"	51	"	"		"		"		"		"		"		"	"			"	"
	"	"	52	"	"		"		"		"		"		"		"	"			"	"

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{G}_1$	1_A	1_Y	2_A	2_Y	3_A	3_Y	GND	4_Y	4_A	5_Y	5_A	6_Y	6_A	$\overline{G}_2$	V_{CC}				
1 T_C = +25°C	V_{OL}	3007	1	0.8 V	2.0 V	32 mA					GND							0.8 V	4.5 V	1Y		0.4	V
		"	2	"			2.0 V	32 mA			"							"	"	2Y		"	"
		"	3	"					2.0 V	32 mA	"							"	"	3Y		"	"
		"	4	"							"	32 mA	2.0 V					"	"	4Y		"	"
		"	5	"							"			32 mA	2.0 V			"	"	5Y		"	"
		"	6	"							"					32 mA	2.0 V	"	"	6Y		"	"
	V_{OH}	3006	7	"	0.8 V	-2 mA					"							"	"	1Y	2.4		"
		"	8	"			0.8 V	-2 mA			"							"	"	2Y	"		"
		"	9	"					0.8 V	-2 mA	"							"	"	3Y	"		"
		"	10	"							"	-2 mA	0.8 V					"	"	4Y	"		"
		"	11	"							"			-2 mA	0.8 V			"	"	5Y	"		"
		"	12	"							"					-2 mA	0.8 V	"	"	6Y	"		"
	I_{O1}		13	2.0 V	2.0 V	2.4 V					"							"	5.5 V	1Y		40	µA
			14	0.8 V	0.8 V	2.4 V					"							2.0 V	"	1Y		"	"
			15	2.0 V			2.0 V	2.4 V			"							0.8 V	"	2Y		"	"
			16	0.8 V			0.8 V	2.4 V			"							2.0 V	"	2Y		"	"
			17	2.0 V					2.0 V	2.4 V	"							0.8 V	"	3Y		"	"
			18	0.8 V					0.8 V	2.4 V	"							2.0 V	"	3Y		"	"
			19	2.0 V							"	2.4 V	2.0 V					0.8 V	"	4Y		"	"
			20	0.8 V							"	2.4 V	0.8 V					2.0 V	"	4Y		"	"
			21	2.0 V							"			2.4 V	2.0 V			0.8 V	"	5Y		"	"
			22	0.8 V							"			2.4 V	0.8 V			2.0 V	"	5Y		"	"
			23	2.0 V							"					2.4 V	2.0 V	0.8 V	"	6Y		"	"
			24	0.8 V							"					2.4 V	0.8 V	2.0 V	"	6Y		"	"
	I_{O2}		25	2.0 V	0.8 V	0.4 V					"							0.8 V	"	1Y		-40	"
			26	0.8 V	2.0 V	0.4 V					"							2.0 V	"	1Y		"	"
			27	2.0 V			0.8 V	0.4 V			"							0.8 V	"	2Y		"	"
			28	0.8 V			2.0 V	0.4 V			"							2.0 V	"	2Y		"	"
			29	2.0 V					0.8 V	0.4 V	"							0.8 V	"	3Y		"	"
			30	0.8 V					2.0 V	0.4 V	"							2.0 V	"	3Y		"	"
			31	2.0 V							"	0.4 V	0.8 V					0.8 V	"	4Y		"	"
			32	0.8 V							"	0.4 V	2.0 V					2.0 V	"	4Y		"	"
			33	2.0 V							"			0.4 V	0.8 V			0.8 V	"	5Y		"	"
			34	0.8 V							"			0.4 V	2.0 V			2.0 V	"	5Y		"	"
			35	2.0 V							"					0.4 V	0.8 V	0.8 V	"	6Y		"	"
			36	0.8 V							"					0.4 V	2.0 V	2.0 V	"	6Y		"	"
	V_{IC}		37		-12 mA						"								4.5 V	1_A		-1.5	V
			38				-12 mA				"								"	2_A		"	"
			39						-12 mA		"								"	3_A		"	"
			40								"		-12 mA						"	4_A		"	"
			41								"				-12 mA				"	5_A		"	"
			42								"						-12 mA		"	6_A		"	"
			43	-12 mA							"								"	$\overline{G}_1$		"	"
			44								"							-12 mA	"	$\overline{G}_2$		"	"
	I_{IH1}	3010	45	2.4 V							"								5.5 V	$\overline{G}_1$		40	µA
		"	46		2.4 V						"								"	1_A		"	"
		"	47				2.4 V				"								"	2_A		"	"
		"	48						2.4 V		"								"	3_A		"	"
		"	49								"		2.4 V						"	4_A		"	"
		"	50								"				2.4 V				"	5_A		"	"
		"	51								"						2.4 V		"	6_A		"	"
		"	52								"							2.4 V	"	$\overline{G}_2$		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1 $\overline{G_1}$	2 1_A	3 1_Y	4 2_A	5 2_Y	6 3_A	7 3_Y	8 GND	9 4_Y	10 4_A	11 5_Y	12 5_A	13 6_Y	14 6_A	15 $\overline{G_2}$	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IH2}	3010	53	5.5 V							GND								5.5 V	$\overline{G_1}$		1	mA
		"	54		5.5 V						"								"	1_A		"	"
		"	55				5.5 V				"								"	2_A		"	"
		"	56						5.5 V		"								"	3_A		"	"
		"	57								"		5.5 V						"	4_A		"	"
		"	58								"				5.5 V				"	5_A		"	"
		"	59								"						5.5 V		"	6_A		"	"
		"	60								"							5.5 V	"	$\overline{G_2}$		"	"
	I_{IL1}	3009	61	2.0 V	0.5 V						"								"				
		"	62	"			0.5 V				"								"	1_A		-40	μA
		"	63	"					0.5 V		"								"	2_A		"	"
		"	64	"							"		0.5 V						"	3_A		"	"
		"	65	"							"				0.5 V				"	4_A		"	"
		"	66	"							"						0.5 V		"	5_A		"	"
																		2.0 V		6_A		"	"
	I_{IL2}	"	67	0.4 V	0.4 V						"								"				
		"	68	"			0.4 V				"								"	1_A		-1.6	mA
		"	69	"					0.4 V		"								"	2_A		"	"
		"	70	"							"		0.4 V						"	3_A		"	"
		"	71	"							"				0.4 V				"	4_A		"	"
		"	72	"							"						0.4 V		"	5_A		"	"
																		0.4 V		6_A		"	"
	I_{IL3}	"	73	"							"								"				
	I_{IL3}		74															0.4 V	"	$\overline{G_1}$		"	"
																				$\overline{G_2}$			
	V_{OC}		75			-12 mA		-12 mA			"								"				
			76							-12 mA	"								"	1_Y		-1.5	V
			77								"	-12 mA							"	2_Y		"	"
			78								"			-12 mA					"	3_Y		"	"
			79								"					-12 mA			"	4_Y		"	"
			80								"								"	5_Y		"	"
																				6_Y			
	I_{OS} *	3011	81	0.8 V	GND	GND					"								"				
		"	82	"			GND	GND	GND	GND	"							0.8 V	"	1_Y	-40	-130	mA
		"	83	"							"	GND	GND					"	"	2_Y	"	"	"
		"	84	"							"			GND	GND			"	"	3_Y	"	"	"
		"	85	"							"					GND		"	"	4_Y	"	"	"
		"	86	"							"						GND	"	"	5_Y	"	"	"
																				6_Y			
	I_{CC}	3005	87	GND	GND		GND		GND		"		GND		GND		GND	GND	"	V_{CC}		77	"
2	Same tests, terminal conditions and limits as for subgroup 1, except V_{IC} and V_{OC} tests are omitted and T_C = +125°C.																						
3	Same tests, terminal conditions and limits as for subgroup 2, except T_C = -55°C.																						
9 T_C = +25°C	t_{PLH}	3003 (Fig. 4)	88	GND	A	OUT					GND							GND	5.0 V	1_A to 1_Y	2	17	ns
		"	89	"			A	OUT	A	OUT	"							"	"	2_A to 2_Y	"	"	"
		"	90	"							"	OUT	A					"	"	3_A to 3_Y	"	"	"
		"	91	"							"			OUT	A			"	"	4_A to 4_Y	"	"	"
		"	92	"							"					OUT	A	"	"	5_A to 5_Y	"	"	"
		"	93	"							"							"	"	6_A to 6_Y	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1 $\overline{G}_1$	2 1_A	3 1_Y	4 2_A	5 2_Y	6 3_A	7 3_Y	8 GND	9 4_Y	10 4_A	11 5_Y	12 5_A	13 6_Y	14 6_A	15 $\overline{G}_2$	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PHL}	3003 (Fig. 4)	94	GND	A	OUT					GND							GND	5.0 V	1_A to 1_Y	2	16	ns
		"	95	"			A	OUT			"							"	"	2_A to 2_Y	"	"	"
		"	96	"					A	OUT	"							"	"	3_A to 3_Y	"	"	"
		"	97	"							"	OUT	A					"	"	4_A to 4_Y	"	"	"
		"	98	"							"			OUT	A			"	"	5_A to 5_Y	"	"	"
		"	99	"							"					OUT	A	"	"	6_A to 6_Y	"	"	"
	t_{PZH}	"	100	B	0.8 V	OUT	0.8 V		0.8 V		"		0.8 V		0.8 V		0.8 V	"	"	$\overline{G}_1$ to 1_Y	"	35	"
	t_{PZL}	"	101	GND	2.0 V	"	2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V	B	"	$\overline{G}_2$ to 1_Y	"	37	"
	t_{PHZ}	"	102	B	0.8 V	"	0.8 V		0.8 V		"		0.8 V		0.8 V		0.8 V	GND	"	$\overline{G}_1$ to 1_Y	"	16	"
	t_{PLZ}	"	103	B	2.0 V	"	2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V	"	"	$\overline{G}_1$ to 1_Y	"	27	"
10 T_C = +125°C	t_{PLH}	"	104	GND	A	"					"							"	"	1_A to 1_Y	"	22	"
		"	105	"			A	OUT			"							"	"	2_A to 2_Y	"	"	"
		"	106	"					A	OUT	"							"	"	3_A to 3_Y	"	"	"
		"	107	"							"	OUT	A					"	"	4_A to 4_Y	"	"	"
		"	108	"							"			OUT	A			"	"	5_A to 5_Y	"	"	"
		"	109	"							"					OUT	A	"	"	6_A to 6_Y	"	"	"
	t_{PHL}	"	110	"	A	OUT					"							"	"	1_A to 1_Y	"	21	"
		"	111	"			A	OUT			"							"	"	2_A to 2_Y	"	"	"
		"	112	"					A	OUT	"							"	"	3_A to 3_Y	"	"	"
		"	113	"							"	OUT	A					"	"	4_A to 4_Y	"	"	"
		"	114	"							"			OUT	A			"	"	5_A to 5_Y	"	"	"
		"	115	"							"					OUT	A	"	"	6_A to 6_Y	"	"	"
	t_{PZH}	"	116	B	0.8 V	OUT	0.8 V		0.8 V		"		0.8 V		0.8 V		0.8 V	"	"	$\overline{G}_1$ to 1_Y	"	40	"
	t_{PZL}	"	117	GND	2.0 V	"	2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V	B	"	$\overline{G}_2$ to 1_Y	"	42	"
	t_{PHZ}	"	118	B	0.8 V	"	0.8 V		0.8 V		"		0.8 V		0.8 V		0.8 V	GND	"	$\overline{G}_1$ to 1_Y	"	20	"
	t_{PLZ}	"	119	B	2.0 V	"	2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V	GND	"	$\overline{G}_1$ to 1_Y	"	32	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

A = Pulse from input pulse gen.

B = Pulse from disable gen.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1 G_1	2 1_A	3 1_Y	4 2_A	5 2_Y	6 3_A	7 3_Y	8 GND	9 4_Y	10 4_A	11 5_Y	12 5_A	13 6_Y	14 6_A	15 $\overline{G}_2$	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	0.8 V	2.0 V	-2 mA					GND								4.5 V	1_Y	2.4		V
		"	2	"			2.0 V	-2 mA			"								"	2_Y	"		"
		"	3	"					2.0 V	-2 mA	"								"	3_Y	"		"
		"	4	"							"	-2 mA	2.0 V						"	4_Y	"		"
		"	5								"			-2 mA	2.0 V			0.8 V	"	5_Y	"		"
		"	6								"					-2 mA	2.0 V	0.8 V	"	6_Y	"		"
	V_{OL}	3007	7	0.8 V	0.8 V	32 mA					"								"	1_Y		0.4	V
		"	8	"			0.8 V	32 mA			"								"	2_Y		"	"
		"	9	"					0.8 V	32 mA	"								"	3_Y		"	"
		"	10								"	32 mA	0.8 V						"	4_Y		"	"
		"	11								"			32 mA	0.8 V			0.8 V	"	5_Y		"	"
		"	12								"					32 mA	0.8 V	0.8 V	"	6_Y		"	"
	I_{O1}		13	2.0 V	2.0 V	2.4 V					"								5.5 V	1_Y		40	µA
			14	"	0.8 V	2.4 V					"								"	1_Y		"	"
			15	"			2.0 V	2.4 V			"								"	2_Y		"	"
			16	"			0.8 V	2.4 V			"								"	2_Y		"	"
			17	"					2.0 V	2.4 V	"								"	3_Y		"	"
			18	"					0.8 V	2.4 V	"								"	3_Y		"	"
			19	"							"	2.4 V	2.0 V						"	4_Y		"	"
			20	"							"	2.4 V	0.8 V						"	4_Y		"	"
			21	"							"			2.4 V	2.0 V			2.0 V	"	5_Y		"	"
			22	"							"			2.4 V	0.8 V			"	"	5_Y		"	"
			23	"							"					2.4 V	2.0 V	"	"	6_Y		"	"
			24	"							"					2.4 V	0.8 V	"	"	6_Y		"	"
	I_{O2}		25	2.0 V	0.8 V	0.4 V					"								"	1_Y		-40	"
			26	"	2.0 V	0.4 V					"								"	1_Y		"	"
			27	"			0.8 V	0.4 V			"								"	2_Y		"	"
			28	"			2.0 V	0.4 V			"								"	2_Y		"	"
			29	"					0.8 V	0.4 V	"								"	3_Y		"	"
			30	"					2.0 V	0.4 V	"								"	3_Y		"	"
			31	"							"	0.4 V	0.8 V						"	4_Y		"	"
			32	"							"	0.4 V	2.0 V						"	4_Y		"	"
			33	"							"			0.4 V	0.8 V			2.0 V	"	5_Y		"	"
			34	"							"			0.4 V	2.0 V			"	"	5_Y		"	"
			35	"							"					0.4 V	0.8 V	"	"	6_Y		"	"
			36	"							"					0.4 V	2.0 V	"	"	6_Y		"	"
	V_{IC}		37		-12 mA						"								4.5 V	1_A		-1.5	V
			38				-12 mA				"								"	2_A		"	"
			39						-12 mA		"								"	3_A		"	"
			40								"		-12 mA						"	4_A		"	"
			41								"				-12 mA				"	5_A		"	"
			42								"						-12 mA		"	6_A		"	"
			43	-12 mA							"								"	$\overline{G}_1$		"	"
			44								"							-12 mA	"	$\overline{G}_2$		"	"
	I_{IH1}	3010	45	2.4 V							"								5.5 V	$\overline{G}_1$		40	µA
		"	46		2.4 V						"								"	1_A		"	"
		"	47				2.4 V				"								"	2_A		"	"
		"	48						2.4 V		"								"	3_A		"	"
		"	49								"		2.4 V						"	4_A		"	"
		"	50								"				2.4 V				"	5_A		"	"
		"	51								"						2.4 V		"	6_A		"	"
		"	52								"							2.4 V	"	$\overline{G}_2$		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{G}_1$	1_A	1_Y	2_A	2_Y	3_A	3_Y	GND	4_Y	4_A	5_Y	5_A	6_Y	6_A	$\overline{G}_2$	V_{CC}				
1 T_C = +25°C	I_{IH2}	3010	53	5.5 V							GND								5.5 V	$\overline{G}_1$		1	mA
		"	54		5.5 V						"								"	1_A		"	"
		"	55				5.5 V				"								"	2_A		"	"
		"	56						5.5 V		"								"	3_A		"	"
		"	57								"		5.5 V						"	4_A		"	"
		"	58								"				5.5 V				"	5_A		"	"
		"	59								"						5.5 V		"	6_A		"	"
		"	60								"							5.5 V	"	$\overline{G}_2$		"	"
	I_{IL1}	3009	61	2.0 V	0.5 V						"								"	1_A		-40	μA
		"	62	"			0.5 V				"								"	2_A		"	"
		"	63	"					0.5 V		"								"	3_A		"	"
		"	64	"							"		0.5 V						"	4_A		"	"
		"	65	"							"				0.5 V				"	5_A		"	"
		"	66	"							"						0.5 V		"	6_A		"	"
	I_{IL2}	"	67	0.4 V	0.4 V						"								"	1_A		-1.6	mA
		"	68	"			0.4 V				"								"	2_A		"	"
		"	69	"					0.4 V		"								"	3_A		"	"
		"	70	"							"		0.4 V						"	4_A		"	"
		"	71	"							"				0.4 V				"	5_A		"	"
		"	72	"							"						0.4 V		"	6_A		"	"
	I_{IL3} I_{IL3}	"	73 74	"							"							0.4 V	"	$\overline{G}_1$ $\overline{G}_2$		"	"
	V_{OC}		75			-12 mA					"								"	1_Y		-1.5	V
			76					-12 mA			"								"	2_Y		"	"
			77							-12 mA	"								"	3_Y		"	"
			78								"	-12 mA							"	4_Y		"	"
			79								"			-12 mA					"	5_Y		"	"
			80								"					-12 mA			"	6_Y		"	"
	I_{OS}	3011	81	0.8 V	5.5 V	GND					"								"	1_Y	-40	-130	mA
		"	82	"			5.5 V	GND			"								"	2_Y	"	"	"
		"	83	"					5.5 V	GND	"								"	3_Y	"	"	"
		"	84	"							"	GND	5.5 V						"	4_Y	"	"	"
		"	85	"							"			GND	5.5 V				"	5_Y	"	"	"
		"	86	"							"					GND	5.5 V		"	6_Y	"	"	"
	I_{CC}	3005	87	GND	GND		GND		GND		"		GND		GND		GND	GND	"	V_{CC}		85	"
2	Same tests, terminal conditions and limits as for subgroup 1, except V_{IC} and V_{OC} tests are omitted and T_C = +125°C.																						
3	Same tests, terminal conditions and limits as for subgroup 2, except T_C = -55°C.																						
9 T_C = +25°C	t_{PLH}	3003 (Fig. 4)	88	GND	A	OUT					GND							GND	5.0 V	1_A to 1_Y	2	16	ns
		"	89	"			A	OUT			"							"	"	2_A to 2_Y	-	"	"
		"	90	"					A	OUT	"							"	"	3_A to 3_Y	"	"	"
		"	91	"							"	OUT	A					"	"	4_A to 4_Y	"	"	"
		"	92	"							"			OUT	A			"	"	5_A to 5_Y	"	"	"
		"	93	"							"					OUT	A	"	"	6_A to 6_Y	"	"	"

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{G}_1$	1_A	1_Y	2_A	2_Y	3_A	3_Y	GND	4_Y	4_A	5_Y	5_A	6_Y	6_A	$\overline{G}_2$	V_{CC}				
9 T_C = +25°C	t_{PHL}	3003 (Fig. 4)	94	GND	A	OUT	A	OUT	A	OUT	GND	OUT	A	OUT	A	OUT	A	GND	5.0 V	1_A to 1_Y	2	22	ns
		"	95	"							"							"	"	2_A to 2_Y	"	"	"
		"	96	"							"							"	"	3_A to 3_Y	"	"	"
		"	97	"							"							"	"	4_A to 4_Y	"	"	"
		"	98	"							"							"	"	5_A to 5_Y	"	"	"
		"	99	"							"							"	"	6_A to 6_Y	"	"	"
	t_{PZH}	"	100	B	2.0 V	OUT	2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V	"	"	$\overline{G}_1$ to 1_Y	"	35	"
	t_{PZL}	"	101	"	0.8 V	"	0.8 V		0.8 V		"		0.8 V		0.8 V		0.8 V	"	"	$\overline{G}_1$ to 1_Y	"	37	"
	t_{PHZ}	"	102	"	2.0 V	"	2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V	"	"	$\overline{G}_1$ to 1_Y	"	16	"
	t_{PLZ}	"	103	"	0.8 V	"	0.8 V		0.8 V		"		0.8 V		0.8 V		0.8 V	"	"	$\overline{G}_1$ to 1_Y	"	27	"
10 T_C = +125°C	t_{PLH}	"	104	GND	A	OUT	A	OUT	A	OUT	"	OUT	A	OUT	A	OUT	A	"	"	1_A to 1_Y	"	21	"
		"	105	"							"							"	"	2_A to 2_Y	"	"	"
		"	106	"							"							"	"	3_A to 3_Y	"	"	"
		"	107	"							"							"	"	4_A to 4_Y	"	"	"
		"	108	"							"							"	"	5_A to 5_Y	"	"	"
		"	109	"							"							"	"	6_A to 6_Y	"	"	"
	t_{PHL}	"	110	"	A	OUT	A	OUT	A	OUT	"	OUT	A	OUT	A	OUT	A	"	"	1_A to 1_Y	"	27	"
		"	111	"							"							"	"	2_A to 2_Y	"	"	"
		"	112	"							"							"	"	3_A to 3_Y	"	"	"
		"	113	"							"							"	"	4_A to 4_Y	"	"	"
		"	114	"							"							"	"	5_A to 5_Y	"	"	"
		"	115	"							"							"	"	6_A to 6_Y	"	"	"
	t_{PZH}	"	116	B	2.0 V	OUT	2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V	"	"	$\overline{G}_1$ to 1_Y	"	40	"
	t_{PZL}	"	117	"	0.8 V	"	0.8 V		0.8 V		"		0.8 V		0.8 V		0.8 V	"	"	$\overline{G}_1$ to 1_Y	"	42	"
	t_{PHZ}	"	118	"	2.0 V	"	2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V	"	"	$\overline{G}_1$ to 1_Y	"	20	"
	t_{PLZ}	"	119	"	0.8 V	"	0.8 V		0.8 V		"		0.8 V		0.8 V		0.8 V	"	"	$\overline{G}_1$ to 1_Y	"	32	"
11	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C.																						

A = Pulse from input pulse gen.

B = Pulse from disable gen.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.8 V, or open).

Subgroup	Symbol*	MIL-STD-883 method	Cases E,F Test no.	1 G_1	2 1_A	3 1_Y	4 2_A	5 2_Y	6 3_A	7 3_Y	8 GND	9 4_Y	10 4_A	11 5_Y	12 5_A	13 6_Y	14 6_A	15 $\overline{G}_2$	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OL}	3007	1	0.8 V	2.0 V	32 mA					GND								4.5 V	1_Y		0.4	V
			2	"			2.0 V	32 mA			"								"	2_Y		"	"
			3	"					2.0 V	32 mA	"								"	3_Y		"	"
			4	"							"	32 mA	2.0 V						"	4_Y		"	"
			5											32 mA	2.0 V			0.8 V		5_Y			
			6													32 mA	2.0 V	0.8 V		6_Y			
	V_{OH}	3006	7	0.8 V	0.8 V	-2 mA					"								"	1_Y	2.4		V
			8	"			0.8 V	-2 mA			"								"	2_Y	"		"
			9	"					0.8 V	-2 mA	"								"	3_Y	"		"
			10								"	-2 mA	0.8 V						"	4_Y	"		"
			11											-2 mA	0.8 V			0.8 V		5_Y			
			12													-2 mA	0.8 V	0.8 V		6_Y			
	I_{O1}		13	2.0 V	2.0 V	2.4 V					"								5.5 V	1_Y		40	μA
			14	"	0.8 V	2.4 V					"								"	1_Y		"	"
			15	"			2.0 V	2.4 V			"								"	2_Y		"	"
			16	"			0.8 V	2.4 V			"								"	2_Y		"	"
			17	"					2.0 V	2.4 V	"								"	3_Y		"	"
			18	"					0.8 V	2.4 V	"								"	3_Y		"	"
			19	"							"	2.4 V	2.0 V						"	4_Y		"	"
			20	"							"	2.4 V	0.8 V						"	4_Y		"	"
			21											2.4 V	2.0 V			2.0 V		5_Y			
			22											2.4 V	0.8 V			"		5_Y			
			23													2.4 V	2.0 V	"		6_Y			
			24													2.4 V	0.8 V	"		6_Y			
	I_{O2}		25	2.0 V	0.8 V	0.4 V					"								"	1_Y		-40	"
			26	"	2.0 V	0.4 V					"								"	1_Y		"	"
			27	"			0.8 V	0.4 V			"								"	2_Y		"	"
			28	"			2.0 V	0.4 V			"								"	2_Y		"	"
			29	"					0.8 V	0.4 V	"								"	3_Y		"	"
			30	"					2.0 V	0.4 V	"								"	3_Y		"	"
			31	"							"	0.4 V	0.8 V						"	4_Y		"	"
			32	"							"	0.4 V	2.0 V						"	4_Y		"	"
			33											0.4 V	0.8 V			2.0 V		5_Y			
			34											0.4 V	2.0 V			"		5_Y			
			35													0.4 V	0.8 V	"		6_Y			
			36													0.4 V	2.0 V	"		6_Y			
	V_{IC}		37		-12 mA						"								4.5 V	1_A		-1.5	V
			38				-12 mA				"								"	2_A		"	"
			39						-12 mA		"								"	3_A		"	"
			40								"		-12 mA						"	4_A		"	"
			41												-12 mA					5_A			
			42														-12 mA			6_A			
			43	-12 mA														-12 mA		$\overline{G}_1$			
			44																	$\overline{G}_2$			
	I_{IH1}	3010	45	2.4 V							"								5.5 V	$\overline{G}_1$		40	μA
			46		2.4 V						"								"	1_A		"	"
			47				2.4 V				"								"	2_A		"	"
			48						2.4 V		"								"	3_A		"	"
			49								"		2.4 V						"	4_A		"	"
			50												2.4 V					5_A			
			51														2.4 V			6_A			
			52															2.4 V		$\overline{G}_2$			

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IH2}	3010	53	5.5 V	5.5 V						GND								5.5 V	$\overline{G}_1$		1	mA
		"	54				5.5 V				"								"	1_A		"	"
		"	55						5.5 V		"								"	2_A		"	"
		"	56								"		5.5 V						"	3_A		"	"
		"	57								"				5.5 V				"	4_A		"	"
		"	58								"						5.5 V		"	5_A		"	"
		"	59								"							5.5 V	"	6_A		"	"
		"	60								"								"	$\overline{G}_2$		"	"
	I_{IL1}	3009	61	2.0 V	0.5 V						"							2.0 V		1_A		-40	μA
		"	62	"			0.5 V				"							"	"	2_A		"	"
		"	63	"					0.5 V		"							"	"	3_A		"	"
		"	64	"							"		0.5 V					"	"	4_A		"	"
		"	65	"							"				0.5 V			"	"	5_A		"	"
		"	66	"							"						0.5 V	"	"	6_A		"	"
	I_{IL2}	"	67	0.4 V	0.4 V						"							0.4 V		1_A		-1.6	mA
		"	68	"			0.4 V				"							"	"	2_A		"	"
		"	69	"					0.4 V		"							"	"	3_A		"	"
		"	70	"							"		0.4 V					"	"	4_A		"	"
		"	71	"							"				0.4 V			"	"	5_A		"	"
		"	72	"							"						0.4 V	"	"	6_A		"	"
	I_{IL3} I_{IL3}	"	73	"							"							0.4 V	"	$\overline{G}_1$ $\overline{G}_2$		"	"
		"	74								"								"			"	"
	V_{OC}		75			-12 mA		-12 mA		-12 mA	"	-12 mA		-12 mA		-12 mA			"	1_Y		-1.5	V
			76								"								"	2_Y		"	"
			77								"								"	3_Y		"	"
			78								"								"	4_Y		"	"
			79								"								"	5_Y		"	"
			80								"								"	6_Y		"	"
	I_{OS}	3011	81	0.8 V	GND	GND	GND	GND	GND	GND	"	GND	GND	GND	GND	GND	GND	0.8 V	"	1_Y	-40	-130	mA
		"	82	"							"							"	"	2_Y	"	"	"
		"	83	"							"							"	"	3_Y	"	"	"
		"	84	"							"							"	"	4_Y	"	"	"
		"	85	"							"							"	"	5_Y	"	"	"
		"	86	"							"							"	"	6_Y	"	"	"
	I_{CC}	3005	87	GND	GND		GND		GND		"		GND		GND		GND	GND	"	V_{CC}		77	"
2	Same tests, terminal conditions and limits as for subgroup 1, except V_{IC} and V_{OC} tests are omitted and T_C = +125°C.																						
3	Same tests, terminal conditions and limits as for subgroup 2, except T_C = -55°C.																						
9 T_C = +25°C	t_{PLH}	3003 (Fig. 4)	88	GND	A	OUT	A	OUT	A	OUT	GND	OUT	A	OUT	A	OUT	A	GND	5.0 V	1_A to 1_Y	2	17	ns
	t_{PHL}	"	89	"							"							"	"	2_A to 2_Y	"	"	"
		"	90	"							"							"	"	3_A to 3_Y	"	"	"
		"	91	"							"							"	"	4_A to 4_Y	"	"	"
		"	92	"							"							"	"	5_A to 5_Y	"	"	"
		"	93	"							"							"	"	6_A to 6_Y	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1 $\overline{G}_1$	2 1_A	3 1_Y	4 2_A	5 2_Y	6 3_A	7 3_Y	8 GND	9 4_Y	10 4_A	11 5_Y	12 5_A	13 6_Y	14 6_A	15 $\overline{G}_2$	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PHL}	3003 (Fig. 4)	94	GND	A	OUT					GND							GND	5.0 V	1_A to 1_Y	2	16	ns
		"	95	"			A	OUT			"							"	"	2_A to 2_Y	"	"	"
		"	96	"					A	OUT	"							"	"	3_A to 3_Y	"	"	"
		"	97	"							"	OUT	A					"	"	4_A to 4_Y	"	"	"
		"	98	"							"			OUT	A			"	"	5_A to 5_Y	"	"	"
		"	99	"							"					OUT	A	"	"	6_A to 6_Y	"	"	"
	t_{PZH}	"	100	B	0.8 V	OUT	0.8 V		0.8 V		"		0.8 V		0.8 V		0.8 V	"	"	$\overline{G}_1$ to 1_Y	"	35	"
	t_{PZL}	"	101	"	2.0 V	"	2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V	"	"	$\overline{G}_1$ to 1_Y	"	37	"
	t_{PHZ}	"	102	"	0.8 V	"	0.8 V		0.8 V		"		0.8 V		0.8 V		0.8 V	"	"	$\overline{G}_1$ to 1_Y	"	16	"
	t_{PLZ}	"	103	"	2.0 V	"	2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V	"	"	$\overline{G}_1$ to 1_Y	"	27	"
10 T_C = +125°C	t_{PLH}	"	104	GND	A	OUT					"							"	"	1_A to 1_Y	"	22	"
		"	105	"			A	OUT			"							"	"	2_A to 2_Y	"	"	"
		"	106	"					A	OUT	"							"	"	3_A to 3_Y	"	"	"
		"	107	"							"	OUT	A					"	"	4_A to 4_Y	"	"	"
		"	108	"							"			OUT	A			"	"	5_A to 5_Y	"	"	"
		"	109	"							"					OUT	A	"	"	6_A to 6_Y	"	"	"
	t_{PHL}	"	110	"	A	OUT					"							"	"	1_A to 1_Y	"	21	"
		"	111	"			A	OUT			"							"	"	2_A to 2_Y	"	"	"
		"	112	"					A	OUT	"							"	"	3_A to 3_Y	"	"	"
		"	113	"							"	OUT	A					"	"	4_A to 4_Y	"	"	"
		"	114	"							"			OUT	A			"	"	5_A to 5_Y	"	"	"
		"	115	"							"					OUT	A	"	"	6_A to 6_Y	"	"	"
	t_{PZH}	"	116	B	0.8 V	OUT	0.8 V		0.8 V		"		0.8 V		0.8 V		0.8 V	"	"	$\overline{G}_1$ to 1_Y	"	40	"
	t_{PZL}	"	117	"	2.0 V	"	2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V	"	"	$\overline{G}_1$ to 1_Y	"	42	"
	t_{PHZ}	"	118	"	0.8 V	"	0.8 V		0.8 V		"		0.8 V		0.8 V		0.8 V	"	"	$\overline{G}_1$ to 1_Y	"	20	"
	t_{PLZ}	"	119	"	2.0 V	"	2.0 V		2.0 V		"		2.0 V		2.0 V		2.0 V	"	"	$\overline{G}_1$ to 1_Y	"	32	"
11	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C.																						

A = Pulse from input pulse gen.

B = Pulse from disable gen.

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d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
V_{IN}	Voltage level at an input terminal.
V_{IC}	Input clamp voltage.
I_{IN}	Current flowing into an input terminal.
T_{PHZ}	Output disable time (of a three-state output) from high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined high level to a high impedance (off) state.
T_{PLZ}	Output disable time (of a three-state output) from low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined low level to a high-impedance (off) state.
T_{PZH}	Output enable time (of a three-state output) to high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined high level.
T_{PZL}	Output enable time (of a three-state output) to low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined low level.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54365
02	54366
03	54367
04	54368

6.6 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuits		
	A	B	C
	Texas Instruments	Signetics Corp.	National Semiconductor Corp.
01	X	X	
02	X	X	
03	X	X	
04	X	X	X

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/163A	Microcircuits, Digital, TTL, Hex Bus Drivers With 3 State Outputs, Monolithic Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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